Overview of Kalman Filter Theory with Applications to Navigation Part III

Michael L. Carroll

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Some Practical Considerations (Numerical)

Topics

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Topics

- Kalman Filter Theory vs. Reality

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- Numerical Stability and Round-Off Errors

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- Numerical Stability and Round-Off Errors
- Matrix Inversion

Kalman Filter Theory vs. Reality

Kalman Assumption Recap

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- Kalman Assumptions

Kalman Filter Theory vs. Reality

Kalman Assumption Recap

- Kalman Assumptions
  - All states linear are modeled correctly
  - Noise parameters are accurately known
  - Initial values are appropriate
  - Process and measurement noise are both Gaussian: $w \sim \mathrm{N}(0, Q)$, and $v \sim \mathrm{N}(0, R)$
  - Process noise and measurement noise are uncorrelated
  - Measurements are taken at regularly scheduled intervals

Kalman Filter Theory vs. Reality

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- But continous filters unrealistic in practice: Always some sort of discretization needed
- Measurements almost never continuous
- Discrete filters acceptable provided continuous estimates are not required
- Suboptimality

Kalman Filter Theory vs. Reality

Sensor Selection

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    Greater number of lower quality sensors can outperform smaller number of higher quality ones (see kalcompare.m)

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  - Maximize the number of directly observable states

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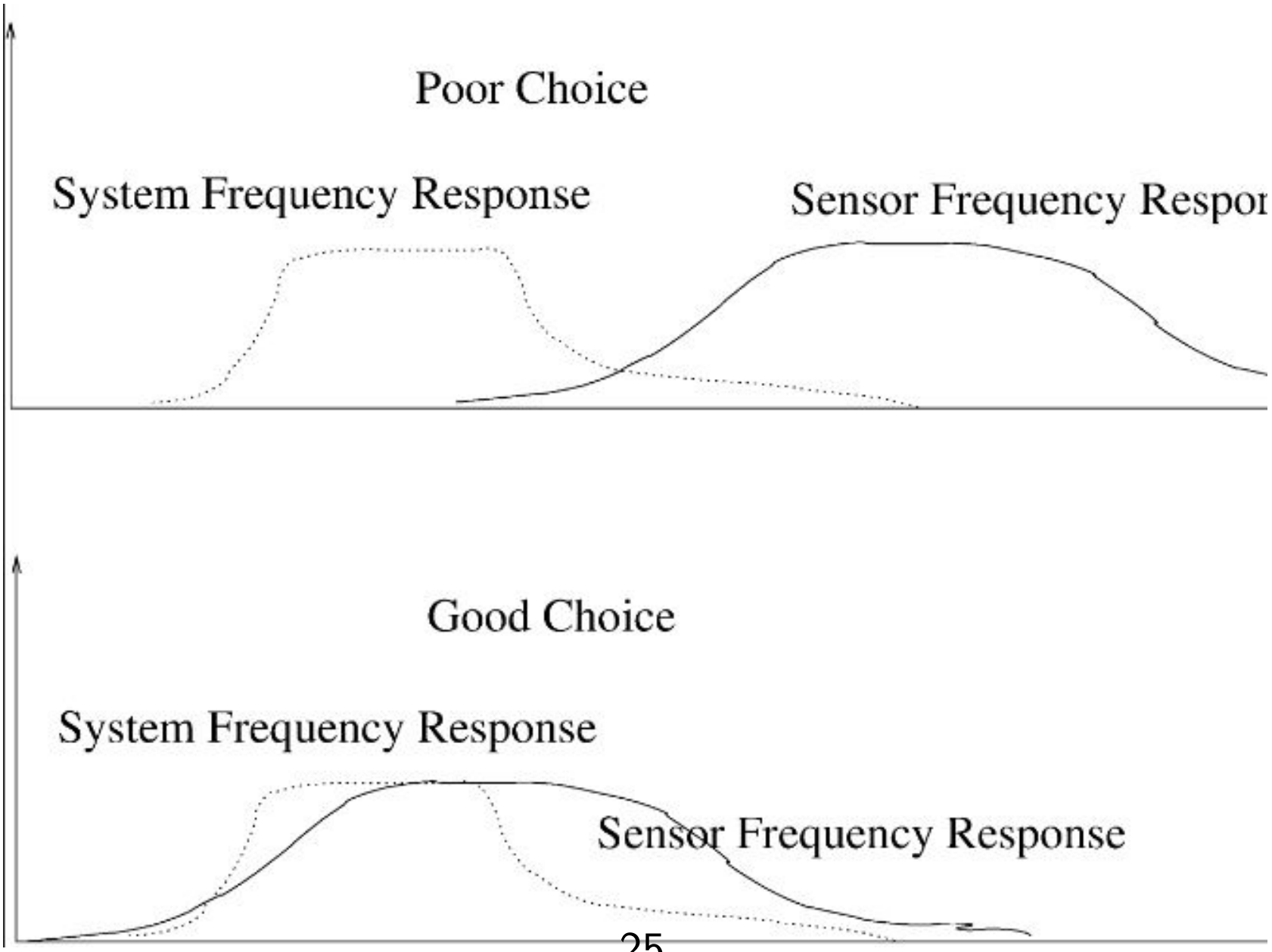
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  - Frequency response of sensor should match that of system
  - Noise spectrum of sensor should be maximally disjoint from that of system


Poor Choice
System Frequency Response
Sensor Frequency Respor
Good Choice
System Frequency Response
Sensor Frequency Response

Kalman Filter Theory vs. Reality

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- Choose sensors to match states.
  E.g., consider the state model

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

  with the measurement model

$$z = \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + v$$

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$$z = \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + v$$

- Here x_1 is observable but x_2 is not, because there are no dynamic paths in $A = I$ flow infomation about x_1 into x_2

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- Examine the H matrix that you would use with this sensor!

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- Examine the H matrix that you would use with this sensor!
- If it renders some states unobservable, then consider adding additional sensors for those states, or replacing this sensor with one that is more favorable

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- Structural observability determined by Φ and H:
- If H is $1 \times n$, where n is the size of the state vector, then the state x_0 is said to be **observable** from n scalar measurements $\{z_0, \ldots, z_{n-1}\}$ if the block matrix

$$\begin{bmatrix} H^\top & \Phi^\top H^\top & \cdots & (\Phi^\top)^{n-1} H^\top \end{bmatrix}$$

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  is of rank n
- Will making measurements as governed by H actually enable us to estimate all of the states in the state vector?

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Designing Observability into the Filter

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- Choosing the right measurement scheduling
- Certain kinds of divergence can be the result of poor observability
- Example: Let $H = [1 \;\; 0 \;\; 0 \;\; 0 \;\; 0]$, $\Phi = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 1 & 2 & 3 & 4 & 5 \end{bmatrix}$

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- Observability challenge for filters is often a matter of choosing right kinds of sensors
- Choosing the right measurement scheduling
- Certain kinds of divergence can be the result of poor observability
- Example: Let $H = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \end{bmatrix}$, $\Phi = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 1 & 2 & 3 & 4 & 5 \end{bmatrix}$
- What is the rank of $\begin{bmatrix} H^\top & \Phi H^\top & \Phi^2 H^\top & \Phi^3 H^\top & \Phi^4 H^\top \end{bmatrix}$?

Filter Divergence

Filter Divergence

- Kinds of Divergence

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- Causes and Remedies

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- Mean estimation error is supposed to converge to 0
  - $\tilde{x}^+(k) = x(k) - \widehat{x}^+(k)$
  - $\mathrm{E}\left[\tilde{x}^+(k)\right] \to 0$
  - Estimate is unbiased: $\mathrm{E}\left[\hat{x}(k)\right] = x(k)$

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  - Estimate is unbiased: $\mathrm{E}\left[\hat{x}(k)\right] = x(k)$
- Covariance matrix for estimation error is same as state estimation error covariance matrix:
  $P^+(k) = \mathrm{E}\left[\tilde{x}^+(k)\left[\tilde{x}^+(k)\right]^\top\right]$

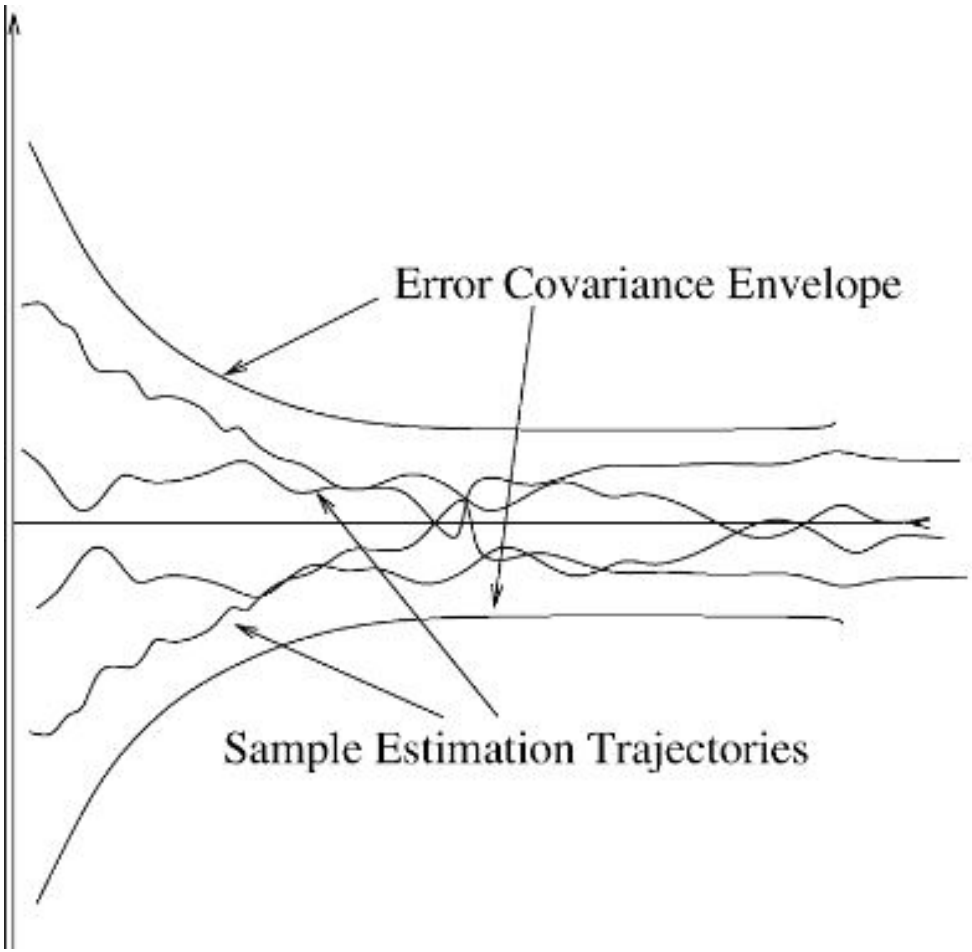
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- For optimal filter, P will indicate convergence or divergence.
  - One common form of divergence due to round-off will send P to zero
- For suboptimal filters, P will not necessarily indicate divergence
  - P may seem to indicate nominal steady-state values and yet the state estimate may go diverge from the true value


Error Covariance Envelope
Sample Estimation Trajectories

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- Loss of symmetry in P
- Loss of positive definiteness: negative numbers on the diagonal!

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- Matrix Conditioning (will address in discussing SRIF)

Round-Off Errors

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Joseph Form

- Computer round-off in floating point arithmetic
- Unit round-off value is largest number ε such that $1 + \varepsilon = 1$
- In Matlab, the global symbol "eps" stores the unit round-off value
- Round-off errors generally plague the classical covariance update algorithm

$$P^{+}(k) = [I - K(k)H(k)]\,P^{-}(k)$$

  - Alternative, called the Joseph form, is more robust:

$$P^{+}(k) = [I - K(k)H(k)]\,P^{-}(k)\,[I - K(k)H(k)]^{\top} + K(k)R(k)K(k)^{\top}$$

Round-Off Errors

De Vries Mechanization

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$$P(+) = P(-) - KHP(-) - P(-)H^\top K^\top + KHP(-)H^\top K^\top + KRK^\top$$

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- Define

$$F = PH^{\top} \tag{1}$$

and

$$D = HF + R. \tag{2}$$

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$$D = HF + R. \tag{2}$$

- We note that D is symmetric, because

$$\begin{aligned} D^\top &= [HF + R]^\top \\ &= [HPH^\top + R]^\top \\ &= [HPH^\top]^\top + R^\top \\ &= [H^\top]^\top P^\top H^\top + R^\top \\ &= HPH^\top + R \qquad \text{(because } P \text{ and } R \text{ are symmetric)} \\ &= HF + R \end{aligned}$$

Round-Off

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- The gain in standard Kalman gain equation

$$K(k) = P^{-}(k)\, H(k)^{\top} \left[H(k) P^{-}(k)\, H(k)^{\top} + R(k) \right]^{-1}$$

can be written as

$$K = FD^{-1}.$$

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- Using Eqs. (1) and (2), we compute

$$\begin{aligned} P(+) &= P(-) - KF^{\top} - FK^{\top} + KDK^{\top} \\ &= P(-) - K \left[F^{\top} - \frac{DK^{\top}}{2} \right] - \left[F - \frac{KD}{2} \right] K^{\top}. \end{aligned}$$

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- Letting $G = F - \frac{KD}{2}$, we obtain

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- The De Vries mechanization implements Eq. (3) by subtracting the ij^{th} element of $GK^{\top}$ from both the ij^{th} and ji^{th} elements of $P(-)$.

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- The De Vries mechanization implements Eq. (3) by subtracting the ij^{th} element of $GK^\top$ from both the ij^{th} and ji^{th} elements of $P(-)$.
- The De Vries form of the covariance update equation is valid for any arbitrary gain matrix K, not just the optimal gains computed from Eqs. (1) and (2).

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- One way to avoid this is to process measurements sequentially as scalars
- Or at least as low-dimensional matrices

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- 1. Let $v = [1 \quad -2]^\top$. Compute the positive definite, symmetrix matrix $M = vv^\top$
- 2. Use the following algorithm (Grewal & Andrews) to factor M into $UDU^\top$ form:

Exercises

```
for j=m:-1:1
    for i=j:-1:1
        covar=P(i,j);
        for k=j+1:m
            covar=covar-U(i,k)*D(k,k)*U(j,k);
        end
        if i==j
            D(j,j)=covar;
            U(j,j)=1;
        else
            U(i,j)=covar/D(j,j);
        end
    end
end
```

Square Root Information Filters

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- Square Root Information Filters (SRIF)

Square Root Information Filters

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- When Λ is "large" P is "small" (in norm)

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- Substitute the Kalman gain formula $K = P^- H^\top \left[HP^- H^\top + R\right]^{-1}$ into the above equation to yield $P^+ = P^- - P^- H^\top \left[HP^- H^\top + R\right]^{-1} HP^-$

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- To futher expand and then simplify, we need to apply the Matrix Inversion Lemma (also known as the Woodbury formula): $(A + BCD)^{-1} = A^{-1} - A^{-1}B\left(C^{-1} + DA^{-1}B\right)^{-1} DA^{-1}$

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- We obtain: $[P^+]^{-1} = [P^-]^{-1} + H^\top R^{-1} H$, or, in terms of Λ: $\Lambda^+ = \Lambda^- + H^\top R^{-1} H$

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- The gain comutation takes on a very simple form: $K = \Lambda^{+} H^{\top} R^{-1}$
- The state is then updated using the gain in the normal way
- Standard KF requires inversion of $r \times r$ matrix (r =number of measurements); information filter requires inversion of a couple of $n \times n$ inversions.
  Thus, if $r >> n$, it is more efficient to use the information filter.

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$$\Lambda^{-}(k) = Q^{-1} - Q^{-1}\Phi\left[\Lambda^{+}(k-1) + \Phi^{\top}Q^{-1}\Phi\right]^{-1}\Phi^{\top}Q^{-1}$$

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- If the process noise covariance Q and/or the measurement noise covariance are constant (or change only infrequently) with time, then the indicated inverses can be computed once at startup or perhaps only occasionaly.

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- Conditioning refers to the condition number of a matrix:
  $\kappa(P) := \sigma_{\text{max}}/\sigma_{\text{min}}$,
  where σ_{max} is the maximum singular value of P and σ_{min} is the minimum singular value.
  Singular values of real-valued P are square roots of eigenvalues of $P^\top P$ (or $PP^\top$)

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- Note that $\kappa(P) \geq 1$, for a nonsingular matrix P, since all of its singular values are positive

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- Example of 2-state model: Highly uncertain first state, but highly certain second state:

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- MATLAB Warning: Matrix is close to singular or badly scaled. Results may be inaccurate. RCOND = 1.000000e-20

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- If we can factor P as the product of a matrix and its transpose, i.e., $P = SS^\top$, then the matrix S is said to be the matrix square root of P

  Note: There are several alternative definitions in the literature $SS^\top, S^2, S^\top S$

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- Key idea: if $P = SS^\top$, then $\kappa(S) = \sqrt{\kappa(P)}$
- Matrix square roots are not unique

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  Note: There are several alternative definitions in the literature
  $SS^\top, S^2, S^\top S$
- Key idea: if $P = SS^\top$, then $\kappa(S) = \sqrt{\kappa(P)}$
- Matrix square roots are not unique
- But all square roots of a matrix are related by orthogonal transformations (i.e., rotations in n-dimensional space)
  T is **orthogonal** if its transpose is its inverse:
  $TT^\top = T^\top T = I$

Square Root Information Filters

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- If $P = S_1 S_1^\top = S_2 S_2^\top$, then there is an orthogonal matrix T such that $S_2 = S_1 T$

Square Root Information Filters

Square Root Filters: Orthogonal Matrices

Square Root Information Filters

Square Root Filters: Orthogonal Matrices

- Important Properties of Orthogonal Matrices:

Square Root Information Filters

Square Root Filters: Orthogonal Matrices

- Important Properties of Orthogonal Matrices:
  - Length Invariance (Isometry): $\|Tx\| = \|x\|$
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    (also preserves dot product between two vectors)

Square Root Information Filters

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Square Root Information Filters

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- Length and Standard Gaussian Invariances play crucial role in square root filtering

Square Root Information Filters

Least Squares Derivation of SRIF (Bierman)

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- As a function of x this becomes

$$J(x) = (z - Hx)^\top (z - Hx)$$

Square Root Information Filters

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$$H^\top H x = H^\top z$$

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- Differentiating $J(x)$ with respect to x (taking into account non-commutative multiplication of matrices), we obtain the least squares **normal equations**:

$$H^\top H x = H^\top z$$

- The least squares solution x_{ls} is therefore

$$x_{\text{ls}} = (H^\top H)^{-1} H^\top z$$

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Least Squares Derivation of SRIF (Bierman)

- Interpreting the errors v statistically and assuming both $\mathrm{E}[v] = 0$ and $\mathrm{E}\left[vv^\top\right] = I$, we derive

$$H^\top H\left(x_{\mathrm{ls}} - x\right) = H^\top v$$

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- Taking the matrix inverse of both sides reveals SRI matrix

$$\Lambda_{\mathrm{ls}} = H^\top H$$

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Square Root Information Filters

Least Squares Derivation of SRIF (Bierman)

- The above derivation was *a posteriori*, i.e., after the measurement
- We can now incorpoate the *a priori* estimate $\hat{x}^-$ and SRI matrix Λ^-
- The previous performance functional is now modified to become

$$J_1(x) := (x - \hat{x}^-)^\top \Lambda^- (x - \hat{x}^-) + (z - Hx)^\top (z - Hx)$$

Square Root Information Filters

Least Squares Derivation of SRIF (Bierman)

Square Root Information Filters

Least Squares Derivation of SRIF (Bierman)

- The vector that minimizes J_1 is denoted $\hat{x}_{\text{ls}}^{+}$ and is easily shown to satisfy

$$(\Lambda^{-} + H^{\top} H)\hat{x}_{\text{ls}}^{+} = \Lambda^{-}\hat{x}^{-} + H^{\top} z$$

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- Finally, the updated covariance is

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- Finally, the updated covariance is

$$P_{\text{ls}}^+ = (\Lambda^- + H^\top H)^{-1}$$

- Bierman points out that Λ^- need not be non-singular
- In fact, one of the advantages of using the information matrix instead of the covariance is that many entries of Λ^- may be 0 (meaning no information=infinite uncerrtainty); we may even start with $\Lambda^- = 0$

Square Root Information Filters

SRIF Recursion (Bierman)

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- Factor the *a priori* Λ^- into square root factors: $\Lambda^- = [S^-]^\top S^-$

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- This allows us to rewrite first term of $J_1(x)$ as

$$(x - \hat{x}^-)^\top \Lambda^- (x - \hat{x}^-) = (z^- - S^- x)^\top (z^- - S^- x)$$

where $z^- = S^- x^-$

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$$(x - \hat{x}^-)^\top \Lambda^- (x - \hat{x}^-) = (z^- - S^- x)^\top (z^- - S^- x)$$

  where $z^- = S^- x^-$
- We can now treat the *a priori* estimate and covariance information as if they were an additional (pseudo) measurement and augment the least squares problem

$$\begin{bmatrix} z^- \\ z \end{bmatrix} = \begin{bmatrix} S^- \\ H \end{bmatrix} x + \begin{bmatrix} v^- \\ v \end{bmatrix}$$

Square Root Information Filters

Summary

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- Results do not depend on which square root is used; facilitates use of orthogonal transformations (Householder, Givens matrices) to put the square roots into numerical convenient forms

Square Root Information Filters

Summary

- Bierman: "The interpretation of the *a priori* as additional observations is the cornerstone of the square root information filter."
- Results do not depend on which square root is used; facilitates use of orthogonal transformations (Householder, Givens matrices) to put the square roots into numerical convenient forms
- Bierman and Thornton (JPL) combined this SRIF formulation with Cholesky decomposition to devise the U-D factorization

UD Factorization

Bierman and Thornton

UD Factorization

Bierman and Thornton

- Based on Cholesky Decomposition

UD Factorization

Bierman and Thornton

- Based on Cholesky Decomposition
- Unit Triangular Matrices: U has 1's on the diagonal and all 0's below the diagonal. E.g.,

$$U = \begin{bmatrix} 1 & 5 & 7 \\ 0 & 1 & 3 \\ 0 & 0 & 1 \end{bmatrix}$$

UD Factorization

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- Factor P^- as $UDU^\top$, where U is unit upper triangular and D is diagonal.

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UD Factorization

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- This last expression is used on the covariance update equation in KH

Overview

- Etymology of the term 'Navigation': Latin *navis*='ship' and *agere*='to move or direct'
- Arose out of marine or ship navigation due to the challenge of navigating without landmarks
- Modern distinction between navigation and guidance
  - navigation associated with state determination (open loop)
  - guidance associated with control feedback (closed loop)

The Navigation Problem from a Mathematical Perspective

With respect to some mathematical reference frame, determine ...

- vector displacement (3d position) and its time derivatives (velocity, acceleration, ...)
- rotational orientation (3d attitude) and its time derivatives (angular velocity, angular acceleration, ...)
- Note: This is a pure mathematics problem ... sensors do not enter into it

The Navigation Problem from a Mathematical Perspective

- Reference frame is crucial:
  - Position: origin and three linearly independent vectors yields three numbers
  - Attitude: rotations about three linearly indepedent vectors yields three numbers
  - I.e., 6 DOF = six degrees of freedom (3 position coordinates + 3 attitude coordinates)

The Navigation Problem from a Real World Physical Perspective

- Which reference frame should we use?
- Rectilinear or curvilenar coordinate system?

The Navigation Problem from a Real World Physical Perspective

Position

For terrestrial navigation, global spherical (spheroidal) coordinates are used:

- Latitude: ± 90 degrees north(+) or south(-) of the equator.
- Longitude: ± 180 degrees east(+) or west(-) of the prime meridian through Greenwich, England
- Height: Above the ellipsoid (normal to surface), or radius from center of earth

The Navigation Problem from a Real World Physical Perspective

Direction: Bearing, Course, Track

- North Reference: True and Magnetic. Direction angles are usually referenced clockwise with respect to north.
- Bearing: The angle with respect to north from one point on earth to another
- Track: The angle made by velocity vector with respect to north
- Course: Sometimes used synonym for track; sometimes expresses desired track vs. actual track
- Heading: Direction vehicle is pointing in with respect to north

The Navigation Problem from a Real World Physical Perspective

The Need for Sensors

- Reference frames are necessary, but not sufficient for real world navigation
- Also need real world sensor measurement data indicative of locations, distances, angles, etc. with respect to these reference frames
- But real world sensors are not perfect! They only provide apparent positions, distances, angles, etc.

The Navigation Problem and Reference Frames

Topics

- Basic Navigation Problem and Solutions
- Earth Referenced Navigation
- Coordinate Frame Conversion
- Basic Strapdown Navigation Equations

Dead Reckoning

- Process of determining new position based on extrapolating from previously known position and velocity.
- Velocity = Speed and Direction
- Unaided inertial navigator is essentially a dead reckoning computer

Earth Referenced Navigation

- Terrestial navigation seeks to answer nav questions with respect to earth
  - Global earth reference
  - Local or relative navigation
- Complications:
  - Earth surface is curved (ellipsoid)
  - Earth is turning (can usually ignore motion about the Sun for terrestrial nav)

Earth Model: WGS 84

- Earth is not spherical, but oblate spheroid or ellipsoid
- Flattened at the poles
- World Geodetic System 84:
  - Semi-major axis: 6,378,137.0 m
  - Semi-minor axis: 6,356,752.314 m
- Not the same as geoid: no terrain or gravity model

Coordinate Frames

- Inertial
- Earth Fixed
- Local Level
- Body

Coordinate Frames

- Technically speaking, all frames are earth-centered
  - Common origen needed for linear transformations (such as rotations)
  - Can compensate by using a translation
- Practically speaking, we think of the Body frame as centered in the body of the vehicle (usually at center of gravity).
- Local Level is thought to be located at surface of earth directly beneath the vehicle.

Inertial Reference Frames

- Non-Rotating
- Non-Accelerating
- Fixed or moving uniformly with respect to another inertial frame (fixed stars)

Inertial Reference Frames

- Nav problem "easy" in inertial frames
- Integrate gyro rates once to get attitude (rotational motion independent of lateral motion)
- Rotate accelerometer outputs to inertial frame using attitude matrix, then integrate to get inertial velocity; integrate velocity to get position.
- Would still have to contend with instrument imperfections.

Inertial Reference Frame Integrations

$$\ddot{r}(t) = a_{\text{sf}} + \text{other external forces} \tag{4}$$

$$v(T) = \dot{r}(T) = v(0) + \int_0^T a_{\text{sf}} dt + \int_0^T (\text{other external forces}) dt \tag{5}$$

$$r(T) = r(0) + \int_0^T v(t) dt \tag{6}$$

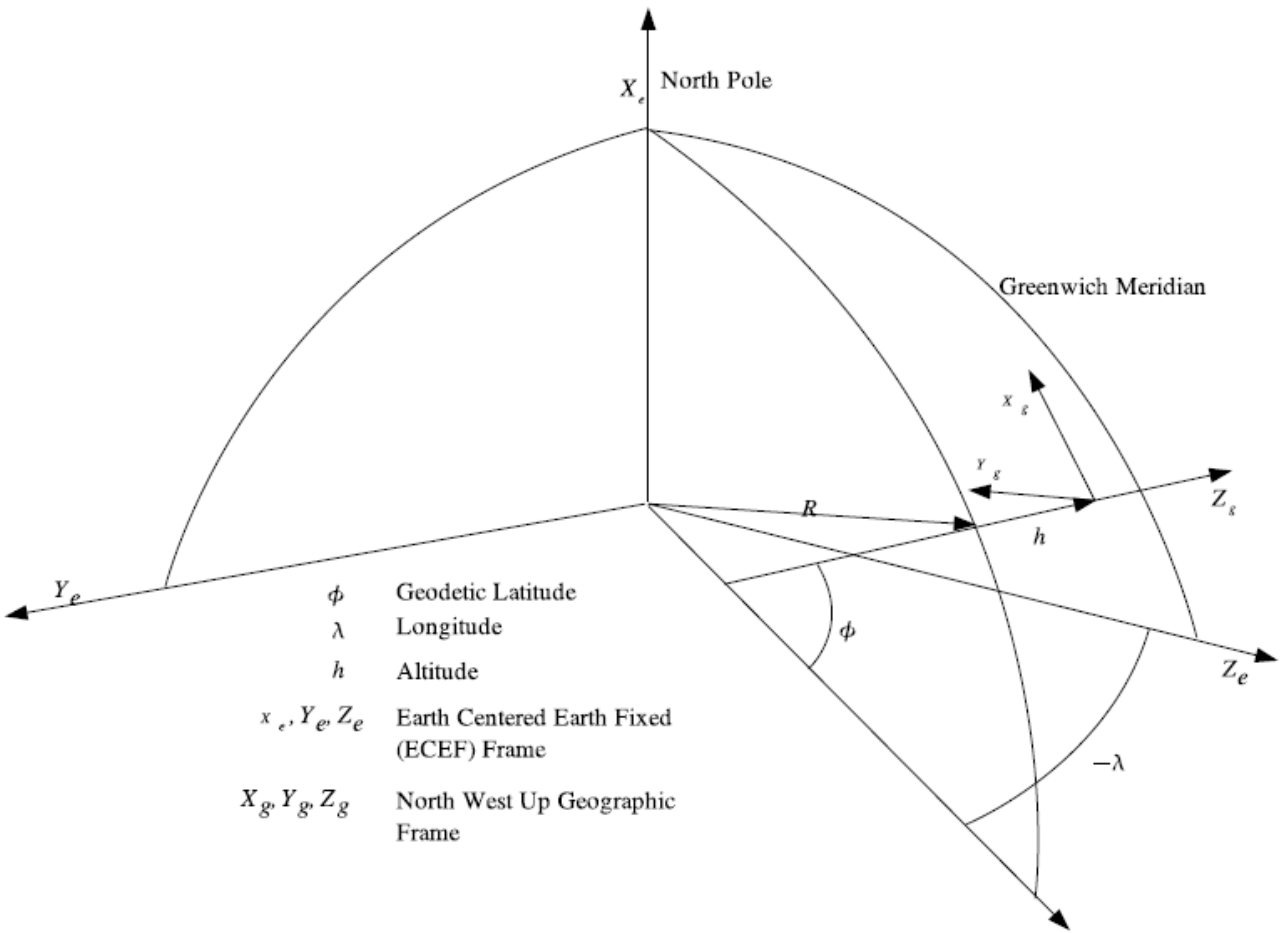
Earth Fixed = ECEF = Earth Centered Earth Fixed

- Rotating frame, i.e., non-inertial
- Cartesian (X, Y, Z in meters)
- Spherical (latitude ϕ, longitude λ, altitude
  - geodetic

Local Level

- Tangent to earth's surface but translated to center of earth.
- Vertical Z-axis is normal to earth's surface
- X, Y-axes are level
- Several kinds: Y north-slaved, wander (0 transport rate) or free azimuth
- Y makes wander angle α with respect to true north (ENU=East, North, Up).

Earth Centered, Earth Fixed: ECEF



Body Frame

- Attached to vehicle
- For aircraft:
  - X out the nose
  - Y out the right wing
  - Z down

Body Frame (cont)

- For space vehicles RIC:
  - R downward on radius to center of earth
  - I in-track, in direction of motion
  - C cross-track, perpendicular to motion
- Also sometimes referred as RAC, where 'A' stands for 'along-track'

Coordinate Frame Conversion

- All frames earth-centered, differing only by rotation
- Rotation represented by direction cosine matrix (DCM) or set of four quaternions
  - Expresses angles made by basis vectors of one frame with those of another
- C_A^B = DCM transformation from frame A to frame B

Coordinate Frame Conversion

- "Direction Cosine" comes from dot product:

$$\langle x, y \rangle = x \cdot y = \|x\| \, \|y\| \cos \theta_{xy}$$

  where θ_{xy} is the angle between the two vectors x and y.
- If x and y are unit vectors, then $\|x\| = \|y\| = 1$ and we have:

$$\langle x, y \rangle = x \cdot y = \cos \theta_{xy}$$

Coordinate Frame Conversion

- If $U = \{u_1, u_2, u_3\}$ and $V = \{v_1, v_2, v_3\}$ are two sets of orthonormal basis vectors, then the DCM relating one frame to the other is given by:

$$C_U^V = \begin{bmatrix} \langle u_1, v_1 \rangle & \langle u_1, v_2 \rangle & \langle u_1, v_3 \rangle \\ \langle u_2, v_1 \rangle & \langle u_2, v_1 \rangle & \langle u_2, v_3 \rangle \\ \langle u_3, v_1 \rangle & \langle u_3, v_2 \rangle & \langle u_3, v_3 \rangle \end{bmatrix}$$

$$= \begin{bmatrix} \cos\theta_{11} & \cos\theta_{12} & \cos\theta_{13} \\ \cos\theta_{21} & \cos\theta_{22} & \cos\theta_{23} \\ \cos\theta_{31} & \cos\theta_{32} & \cos\theta_{33} \end{bmatrix}$$

Coordinate Frame Conversion

- If $V = \{v_1, v_2, v_3\}$ is considered to be the reference frame and its units vectors are the standard basis so that

$$v_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, v_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, v_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix},$$

then the DCM simplifies to just the coordinates of u_1, u_2, u_3 vectors with respect to V:

$$C_U^V = \begin{bmatrix} u_{11} & u_{12} & u_{13} \\ u_{21} & u_{22} & u_{23} \\ u_{31} & u_{32} & u_{33} \end{bmatrix}$$

Coordinate Frame Conversion

- DCMs are (real) orthogonal matrices
  $\Rightarrow \left[C_A^B\right]^{-1} = \left[C_A^B\right]^\top = C_B^A$
- Third row (or column) is vector cross product of other two
  $\Rightarrow$ only six independent parameters
- DCMs satisfy: $C_B^A C_C^B = C_C^A$

Coordinate Frame Conversion

- Orthogonal matrices preserve lengths of vectors: $\|Rx\| = \|x\|$
- Orthogonal matrices also preserve angles between vectors: $\langle Rx, Ry \rangle = \left\langle R^\top Rx, y \right\rangle = \langle Ix, y \rangle = \langle x, y \rangle$
- Intuitively obvious in the case of rotations

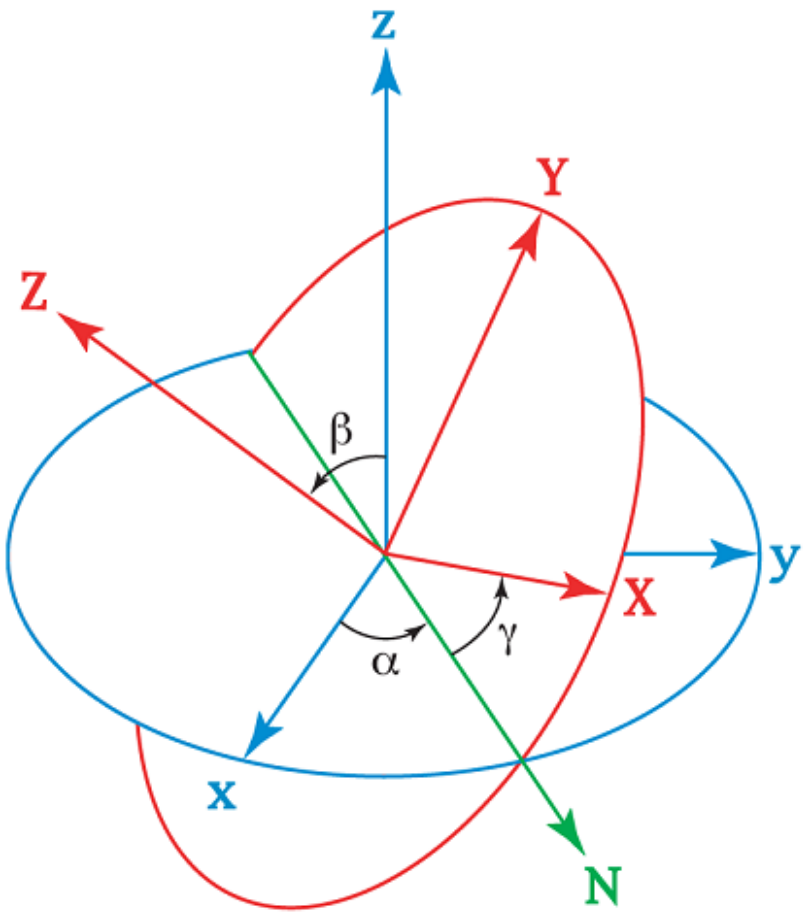
DCMs and Euler Rotation

- Euler showed that three simultaneous rotations about three orthogonal axes can be viewed as a single rotation about a single axis
- The single angle θ of rotation for a DCM can be determined from the eigenvalues of the DCM, which are:

$$\{1, \exp(i\theta), \exp(-i\theta)\}$$

- The determinant of a DCM is therefore always 1. Why? (Hint: How do you compute the determinant from the eigenvalues?)

Euler Rotation



DCMs and Euler Rotation

- The axis of rotation for the single Euler rotation can be found from DCM as follows:

$$C_U^V = \begin{bmatrix} u_{11} & u_{12} & u_{13} \\ u_{21} & u_{22} & u_{23} \\ u_{31} & u_{32} & u_{33} \end{bmatrix}$$

then the axis of rotation $r = [r_1, r_2, r_3]^\top$ is given by

$$\begin{aligned} r_1 &= u_{12}u_{23} - (u_{22} - 1)u_{13} \\ r_2 &= u_{21}u_{13} - (u_{11} - 1)u_{23} \\ r_3 &= (u_{11} - 1)(u_{22} - 1) - u_{12}u_{21} \end{aligned}$$

DCMs and Quaternions

- If C is a direction cosine matrix with elements c_{ij}, then the following four quantities define a quaternion of the rotation:

$$q_1 = \frac{1}{4q_4}(c_{23} - c_{32})$$
$$q_2 = \frac{1}{4q_4}(c_{13} - c_{31})$$
$$q_3 = \frac{1}{4q_4}(c_{21} - c_{12})$$
$$q_4 = \pm\frac{1}{2}\sqrt{1 + c_{11} + c_{22} + c_{33}}$$

DCMs and Quaternions

- It's even simpler to obtain quaternions from Euler rotation angle θ and axis given by a unit vector u:

$$q_i = u_i \sin(\theta/2), \text{ for } i = 1, 2, 3$$

and

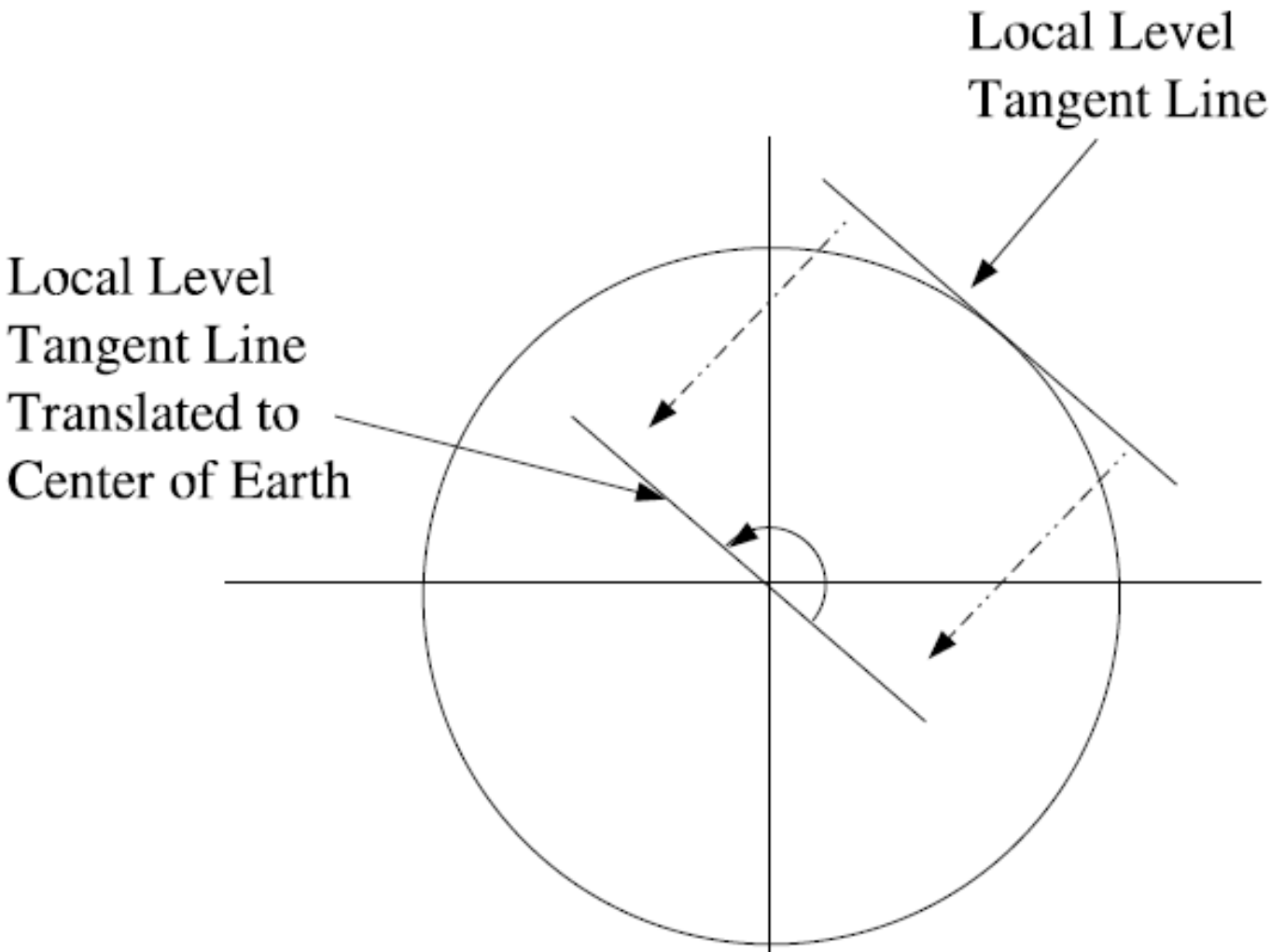
$$q_4 = \cos(\theta/2)$$

Coordinate Frame Conversion: Most important DCMs for terrestrial navigation

- Body to Local Level: C_B^L
  - sometimes called Body to Nav and notated C_B^N
  - yields attitude info (Euler angles for roll, pitch, yaw)
- Local Level to Earth: C_L^E
  - yields position info and also used to transform earth rate to local level

Coordinate Frame Conversion: Most important DCMs

- Visualize C_L^E's determination of position as follows:
  - As vehicle moves, local level frame changes its orienation with respect to earth's center in order to maintain local level with respect to earth's surface
  - If you think of local level frame as attached to earth's surface, you get distracted by the displacement of the frame
  - If you move local level frame to center of earth, the displacement disappears and all that remains are the rotations


Local Level
Tangent Line
Local Level
Tangent Line
Translated to
Center of Earth

- Gimbaled INS:
  - Gyro stabilized platform
  - Gimbal assembly and torque motors
  - Resolvers, slip rings, bearings, etc. to control platform
  - Goal: Maintain stable, locally level or inertial reference frame

Strapdown Navigation: Strapdown vs. Gimbaled

- Gimbaled INS Issues
  - Reliability and Cost: Lots of moving parts!
  - Body rate and acceleration information not directly available
  - Rate signals are generally noisy and have low bandwidth

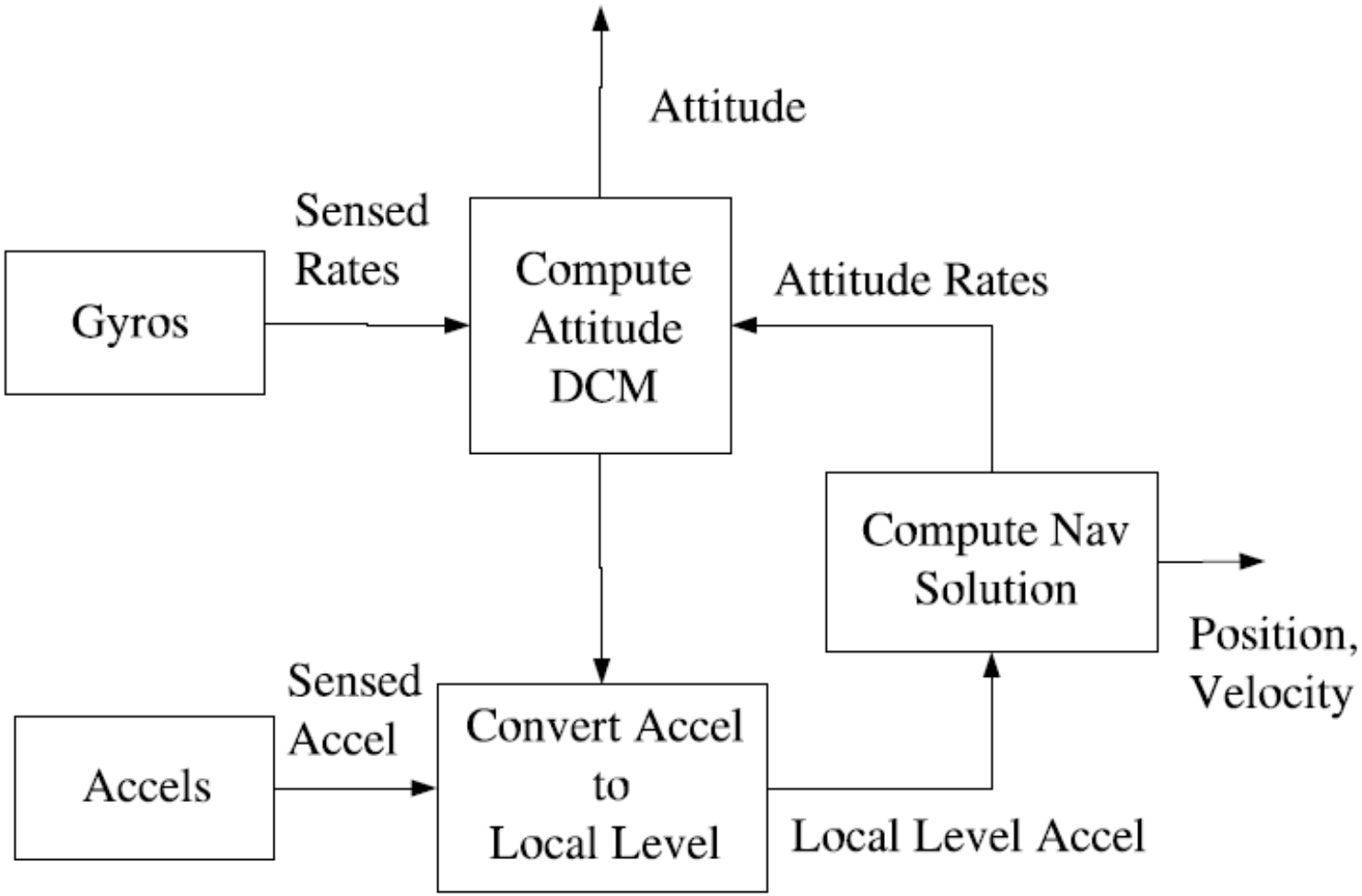
Strapdown Navigation: Strapdown vs. Gimbaled

- Strapdown INS:
  - Inertial instruments strapped to body frame
  - Requires local level accelerations to be computed using direction cosine matrix C_B^L
  - C_B^L derived from gyro-sensed body rates
  - Strapdown systems can be implemented with either rate gyros or attitude gyros

Strapdown Navigation: Strapdown vs. Gimbaled

- Strapdown INS Advantages
  - Immediate detection of body rates and accelerations
  - Fewer moving parts
  - Potential redundancy advantages
- Gimbaled systems are nearly obsolete

Strapdown Inertial Block Diagram



Basic Strapdown Navigation Equations

- Need to compute attitude, velocity, and position
- I.e., need to compute C_B^L and C_L^E
- These DCMs are dynamic quantities
- C_B^L determines attitude and can be rapidly varying
- C_L^E determines position and is slowly varying (relative to attitude dynamics)

Strapdown Navigation Equations

- Three differential equations
  - Equation for body to local level transformation C_B^L
  - Equation for velocity (uses C_B^L)
  - Equation for local level to earth for position DCM C_L^E

Strapdown Navigation Equations

- Two Algebraic Equations
  - Equation for converting velocity to transport rate (to be explained)
  - Equation for converting earth rate to local level coordinates

$$\dot{C}_B^L = C_B^L \Omega_{IB}^B - \left(\Omega_{IE}^L + \Omega_{EL}^L\right) C_B^L \tag{7}$$

$$\dot{v}^L = C_B^L a_{\text{sf}}^B + g' - \left(2\Omega_{IE}^L + \Omega_{EL}^L\right) v^L \tag{8}$$

$$\dot{C}_L^E = C_L^E \Omega_{EL}^L \tag{9}$$

$$\omega_{EL}^{L} = \frac{1}{r} u_r^L \times v^L \text{ (wander azimuth } \rho_z = 0) \tag{10}$$

$$\omega_{IE}^{L} = \left[C_L^E \right]^\top \omega_{IE}^{E} \tag{11}$$

Strapdown Navigation Equations

The C_B^L Equation (7)

$$\dot{C}_B^L = C_B^L \Omega_{IB}^B - \left(\Omega_{IE}^L + \Omega_{EL}^L\right) C_B^L$$

- All the Ω matrices represent rotation vector cross products in skew-symmetric matrix format
  - Ω_{IB}^B is rotational rate of aircraft in body coordinates
  - Ω_{IE}^L is earth rate in local level coordinates
  - Ω_{EL}^L is transport rate in local level coordinates

Strapdown Navigation Equations: Detour into Cross Product Matrices

- The cross product of $\omega = \begin{bmatrix} \omega_1 \\ \omega_2 \\ \omega_3 \end{bmatrix}$ with any other vector v is the same as multiplying v on the left by $\Omega = \begin{bmatrix} 0 & -\omega_3 & \omega_2 \\ \omega_3 & 0 & -\omega_1 \\ -\omega_2 & \omega_1 & 0 \end{bmatrix}$
- $\omega \times v = \Omega v$

Strapdown Navigation Equations

The C_B^L Equation (7)

$$\dot{C}_B^L = C_B^L \Omega_{IB}^B - \left(\Omega_{IE}^L + \Omega_{EL}^L\right) C_B^L$$

- Transport rate is the rotational rate of the aircraft due to motion over the surface of the earth. Is a function of velocity.
- Ω_{IB}^B is total inertial angular rate sensed by gyros and includes effects of transport and earth rates.
- Hence transport and earth rate effects have to be subtracted off.

Strapdown Navigation Equations

The Strapdown Velocity Equation

(8)

$$\dot{v}^L = C_B^L a_{\text{sf}}^B + g' - \left(2\Omega_{IE}^L + \Omega_{EL}^L\right) v^L$$

- Sometimes referred to as the velocity rate equation
- a_{sf}^B is specific force sensed by accelerometers in body coordinates (because the accels are strapped to the body – hence the term strapdown)

Strapdown Navigation Equations

The Transport Rate Equation

(9)
$$\dot{C}_L^E = C_L^E \Omega_{EL}^L$$

- Connected to velocity rate equation by the first algebraic equation (10):
$$\omega_{EL}^L = \frac{1}{r} u_r^L \times v^L$$
- Eq. (9) Sometimes referred to as position rate equation, because C_L^E directly implies the vehicle position

Strapdown Navigation Equations

Position Rate / Transport Rate Equations (Geographic)

- North transport rate:

$$\rho_N = \frac{v_E}{R_e}\left[1 - \frac{h}{R_e} - e\sin^2\phi\right] \tag{12}$$

- East transport rate:

$$\rho_E = -\frac{v_N}{R_e}\left[1 - \frac{h}{R_e} - e\left(1 - 3\cos^2\phi\right)\right] \tag{13}$$

where e is ellipticity (approx. $\frac{1}{298}$) and R_e is mean equatorial earth radius

Strapdown Navigation Equations

Position Rate / Transport Rate Equations (Geographic)

- Latitude Rate:

$$\dot{\phi} = -\rho_E \tag{14}$$

- Longitude Rate:

$$\dot{\lambda} = \rho_N \sec \phi \tag{15}$$

Strapdown Navigation Equations

- Geographic (north-slaved) mechanization has singularities at poles
- Two other approaches to local level coordinates:
  - Free azimuth: vertical component of transport rate $\rho_z =$ additive inverse of vertical component of earth rate
  - Wander Azimuth: vertical component of transport rate $\rho_z = 0$

Strapdown Navigation Equations

Position Rate / Transport Rate Equations (Wander and Free Azimuth)

- X Transport Rate:

$$\rho_x = -\frac{v_y}{R_e}\left[1 - \frac{h}{R_e} - e\left(1 - 3d_{22}^2 - d_{21}^2\right)\right] - \frac{v_x}{R_e}\left(2d_{21}d_{22}\right)e \tag{16}$$

- Y Transport Rate:

$$\rho_y = \frac{v_x}{R_e}\left[1 - \frac{h}{R_e} - e\left(1 - 3d_{21}^2 - d_{22}^2\right)\right] + \frac{v_y}{R_e}\left(2d_{21}d_{22}\right)e \tag{17}$$

Strapdown Navigation Equations

Position Rate / Transport Rate Equations (Wander and Free Azimuth)

- Azimuth Rate:

$$\rho_z = \begin{cases} -\Omega_e d_{23}, & \text{if free azimuth;} \\ 0, & \text{if wander azimuth} \end{cases}$$

- In equations above, the matrix $[d_{ij}]$ is the direction cosine matrix C_L^E from local level (L) frame to earth centered earth fixed (E) frame.

GPS

- Trilateration: Knowing the distance from an unknown point to three known points yields the coordinates of the unknown point
- What if the distances are not known precisely? Answer: Use 'pseudorange' instead, i.e., use signals traveling at speed of light and very accurate clocks
- What if the coordinates of the three points are not really known precisely? Answer: Use your best guess!

Trilateration

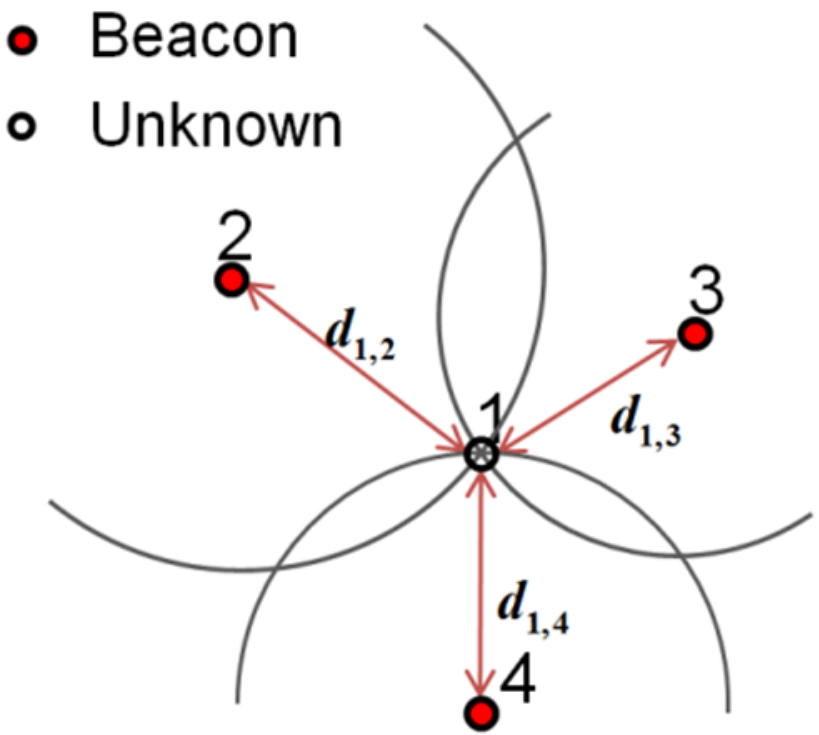


Figure: Trilateration: Distances to Three Beacons

Pseudorange

- distance = (speed of signal) $\times$ (time of signal flight)
  - $\rho = c\Delta t$
- Called 'pseudorange' because it is not a true range; it is only apparent due to inaccurate clocks
- By the same token, the coordinates of the transmitting satellites are also not known precisely, and so we could speak of 'pseudopositions' as well. But nobody does that!

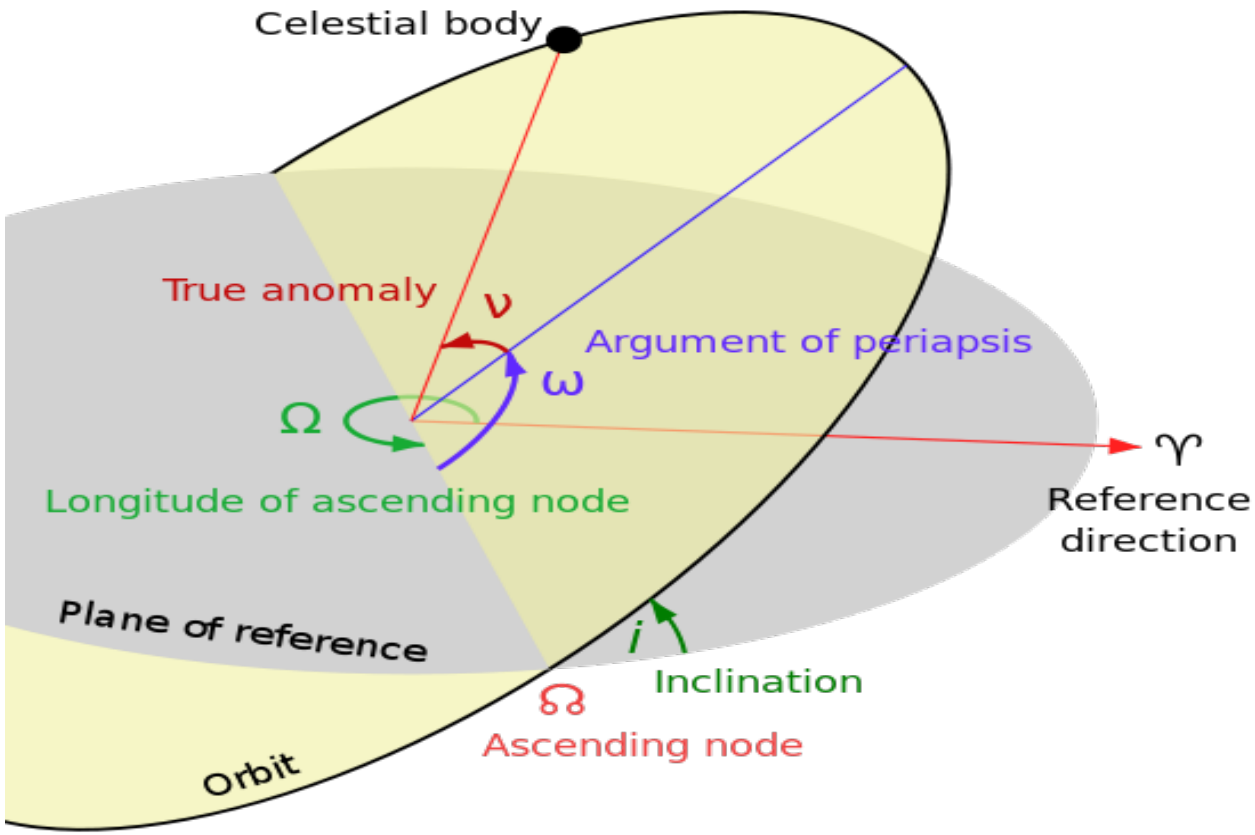
Satellite Position

- How do we know which satellite is transmitting? Answer: Unique Pseudorandom Noise (PRN)
- How we know the position of the satellite at time of transmission? Answer:
  - Six Keplerian elements
  - User receiver converts to WGS-84 X,Y,Z ECEF coordinates

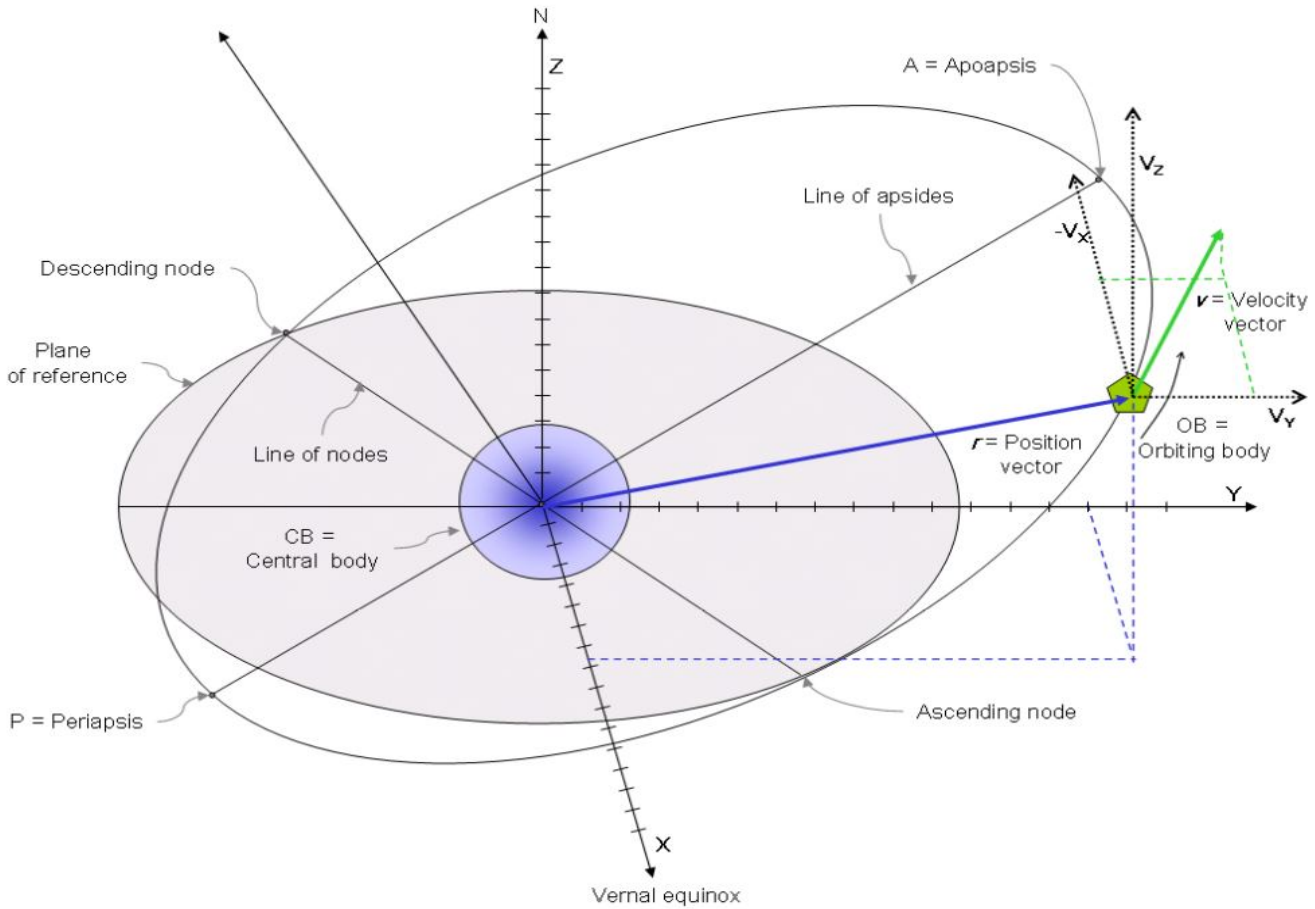
Six Keplerian Elements

- Eccentricity
- Semi-major axis
- Inclination
- Longitude of the ascending node
- Argument of periaspsis
- Mean anomaly

Keplerian Elements



Keplerian Elements



IS-GPS-200 Table 20-II

1 of 2

Table 20-II. Ephemeris Data Definitions	
M_0	Mean Anomaly at Reference Time
Δn	Mean Motion Difference From Computed Value
e	Eccentricity
$\sqrt{A}$	Square Root of the Semi-Major Axis
Ω_0	Longitude of Ascending Node of Orbit Plane at Weekly Epoch
i_0	Inclination Angle at Reference Time
ω	Argument of Perigee
$\dot{\Omega}$	Rate of Right Ascension
IDOT	Rate of Inclination Angle

IS-GPS-200 Table 20-II

2 of 2

C_{uc}	Amplitude of the Cosine Harmonic Correction Term to the Argument of Latitude
C_{us}	Amplitude of the Sine Harmonic Correction Term to the Argument of Latitude
C_{rc}	Amplitude of the Cosine Harmonic Correction Term to the Orbit Radius
C_{rs}	Amplitude of the Sine Harmonic Correction Term to the Orbit Radius
C_{ic}	Amplitude of the Cosine Harmonic Correction Term to the Angle of Inclination
C_{is}	Amplitude of the Sine Harmonic Correction Term to the Angle of Inclination
t_{oe}	Reference Time Ephemeris (reference paragraph 20.3.4.5)
IODE	Issue of Data (Ephemeris)

IS-GPS-200 Table 20-IV

Elements of Coordinate Systems (1 of 3)

Table 20-IV. Elements of Coordinate Systems (sheet 1 of 2)	
$\mu = 3.986005 \times 10^{14}$ meters^3/sec^2	WGS 84 value of the earth's gravitational constant for GPS user
$\dot{\Omega}_e = 7.2921151467 \times 10^{-5}$ rad/sec	WGS 84 value of the earth's rotation rate
$A = \left(\sqrt{A}\right)^2$	Semi-major axis
$n_0 = \sqrt{\frac{\mu}{A^3}}$	Computed mean motion (rad/sec)
$t_k = t - t_{oe}$*	Time from ephemeris reference epoch
$n = n_0 + \Delta n$	Corrected mean motion

IS-GPS-200 Table 20-IV

Elements of Coordinate Systems (2 of 3)

$M_k = M_0 + nt_k$	Mean anomaly
$M_k = E_k - e \sin E_k$	Kepler's Equation for Eccentric Anomaly (may be solved by iteration) (radians)
$\nu_k = \tan^{-1}\left\{\frac{\sin \nu_k}{\cos \nu_k}\right\}$ $= \tan^{-1}\left\{\frac{\sqrt{1-e^2}\sin E_k / (1-e\cos E_k)}{(\cos E_k - e)/(1-e\cos E_k)}\right\}$	True Anomaly

* t is GPS system time at time of transmission, i.e., GPS time corrected for transit time (range/speed of light). Furthermore, t_k shall be the actual total time difference between the time t and the epoch time t_{oe}, and must account for beginning or end of week crossovers. That is, if t_k is greater than 302,400 seconds, subtract 604,800 seconds from t_k. If t_k is less than -302,400 seconds, add 604,800 seconds to t_k.

IS-GPS-200 Table 20-IV

Elements of Coordinate Systems (3 of 3)

Table 20-IV. Elements of Coordinate Systems (sheet 2 of 2)

$E_k = \cos^{-1}\left\{\frac{e + \cos \nu_k}{1 + e \cos \nu_k}\right\}$		Eccentric Anomaly
$\Phi_k = \nu_k + \omega$		Argument of Latitude
$\delta u_k = c_{us}\sin 2\Phi_k + c_{uc}\cos 2\Phi_k$	Argument of Latitude Correction	Second Harmonic Perturbations
$\delta r_k = c_{rs}\sin 2\Phi_k + c_{rc}\cos 2\Phi_k$	Radius Correction	
$\delta i_k = c_{is}\sin 2\Phi_k + c_{ic}\cos 2\Phi_k$	Inclination Correction	
$u_k = \Phi_k + \delta u_k$		Corrected Argument of Latitude
$r_k = A(1 - e \cos E_k) + \delta r_k$		Corrected Radius
$i_k = i_0 + \delta i_k + (\mathrm{IDOT})\, t_k$		Corrected Inclination
$x_k' = r_k \cos u_k$ $y_k' = r_k \sin u_k$		Positions in orbital plane.
$\Omega_k = \Omega_0 + (\dot{\Omega} - \dot{\Omega}_e)\, t_k - \dot{\Omega}_e\, t_{oe}$		Corrected longitude of ascending node.
$x_k = x_k' \cos\Omega_k - y_k' \cos i_k \sin\Omega_k$ $y_k = x_k' \sin\Omega_k + y_k' \cos i_k \cos\Omega_k$ $z_k = y_k' \sin i_k$		Earth-fixed coordinates.

Inertial Error Dynamics

Topics

Inertial Error Dynamics

Topics

- Imperfect Inertial Instruments

Inertial Error Dynamics

Topics

- Imperfect Inertial Instruments
- From Strapdown Equations to Error Equations

Inertial Error Dynamics

Topics

- Imperfect Inertial Instruments
- From Strapdown Equations to Error Equations
- Initialization: Alignment

Inertial Error Dynamics

Imperfect Inertial Instruments

Inertial Error Dynamics

Imperfect Inertial Instruments

- Accelerometers and Gyros have inherent imperfections

Inertial Error Dynamics

Imperfect Inertial Instruments

- Accelerometers and Gyros have inherent imperfections
- Bias, scale factor, g^2 terms, noise, random walks, etc.

Inertial Error Dynamics

Imperfect Inertial Instruments

- Accelerometers and Gyros have inherent imperfections
- Bias, scale factor, g^2 terms, noise, random walks, etc.
- Inertial sensor clusters also have errors: misalignments

Inertial Error Dynamics

Imperfect Inertial Instruments

- Accelerometers and Gyros have inherent imperfections
- Bias, scale factor, g^2 terms, noise, random walks, etc.
- Inertial sensor clusters also have errors: misalignments
- Many error sources are calibrated out using rate tables during ATP

Inertial Error Dynamics

Imperfect Inertial Instruments

- Accelerometers and Gyros have inherent imperfections
- Bias, scale factor, g^2 terms, noise, random walks, etc.
- Inertial sensor clusters also have errors: misalignments
- Many error sources are calibrated out using rate tables during ATP
- Residual effects will influence strapdown nav equations

Inertial Error Dynamics

Imperfect Inertial Instruments

Inertial Error Dynamics

Imperfect Inertial Instruments

- Strapdown nav equations very sensitive to gyro errors

Inertial Error Dynamics

Imperfect Inertial Instruments

- Strapdown nav equations very sensitive to gyro errors
- Gyro biases yield erroneous C_B^L matrix which in turn causes erroneous accelerations to be computed in local level frame

Inertial Error Dynamics

Imperfect Inertial Instruments

- Strapdown nav equations very sensitive to gyro errors
- Gyro biases yield erroneous C_B^L matrix which in turn causes erroneous accelerations to be computed in local level frame
- Thus, gyro errors affect not only attitude, but also position and velocity

Inertial Error Dynamics

Simple Inertial Error Models

- Gyros

$$\epsilon\omega_x = \Delta B_{gx} + \Delta K_{gx}\omega_x + \theta_{gxy}\omega_y + \theta_{gxz}\omega_z$$

$$\epsilon\omega_y = \Delta B_{gy} + \Delta K_{gy}\omega_y + \theta_{gyz}\omega_z + \theta_{gyx}\omega_x$$

$$\epsilon\omega_z = \Delta B_{gz} + \Delta K_{gz}\omega_z + \theta_{gzx}\omega_x + \theta_{gzy}\omega_y$$

where ΔB_{gi} is the ith gyro bias, ΔK_{gi} the ith gyro scale factor, and θ_{gij} is the misalignment of the ith gyro axis toward the jth gyro axis.

Inertial Error Dynamics

Simple Inertial Error Models

- Accelerometers

$$\epsilon a_x = \Delta B_{ax} + \Delta K_{ax} a_x + \theta_{axy} a_y + \theta_{axz} a_z$$

$$\epsilon a_y = \Delta B_{ay} + \Delta K_{ay} a_y + \theta_{ayz} a_z + \theta_{ayx} a_x$$

$$\epsilon a_z = \Delta B_{az} + \Delta K_{az} a_z + \theta_{azx} a_x + \theta_{azy} a_y$$

where ΔB_{ai} is the ith accelerometer bias, ΔK_{ai} the ith accelerometer scale factor, and θ_{aij} is the misalignment of the ith accelerometer axis toward the jth accelerometer axis.

Inertial Error Dynamics

Bias Models

Inertial Error Dynamics

Bias Models

- Bias in both types of instruments can be further broken down into several components:
  - fixed bias – constant from turn-on to turn-on – should be calibrated out
  - bias stability – varies from turn-on to turn-on – can't be calibrated out, but can be characterized in terms of variance – can model as random constant
  - bias drift – usually modeled as random walk

Inertial Error Dynamics

Scale Factor Models

Inertial Error Dynamics

Scale Factor Models

- Scale factor can also be modeled as a sum of three components

Inertial Error Dynamics

Scale Factor Models

- Scale factor can also be modeled as a sum of three components
- Often in nav filters, scale factor residual is modeled as a random constant

Inertial Error Dynamics

Scale Factor Models

- Scale factor can also be modeled as a sum of three components
- Often in nav filters, scale factor residual is modeled as a random constant
- Important to keep in mind that the contribution of scale factor error is a function of input (rate or acceleration)

Inertial Error Dynamics

Higher Order Models

Inertial Error Dynamics

Higher Order Models

- In expanding the output error as a Taylor series function of input, bias, scale factor, and misalignment show up as zero and first order terms.

Inertial Error Dynamics

Higher Order Models

- In expanding the output error as a Taylor series function of input, bias, scale factor, and misalignment show up as zero and first order terms.
- Higher order terms are also present, especially second order terms.

Inertial Error Dynamics

Higher Order Models

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- Higher order terms are also present, especially second order terms.
- In some calibration and alignment filters, such higher order effects are modelled

Inertial Error Dynamics

Higher Order Models

- In expanding the output error as a Taylor series function of input, bias, scale factor, and misalignment show up as zero and first order terms.
- Higher order terms are also present, especially second order terms.
- In some calibration and alignment filters, such higher order effects are modelled
- SNU-84-1 includes error budgets associated with all the combined, higher order effects.

Inertial Error Dynamics

Higher Order Models: Taylor Series of Two Variables

Inertial Error Dynamics

Higher Order Models: Taylor Series of Two Variables

$$
\begin{aligned}
f(x,y) = f(x_0,y_0) &+ f_x(x_0,y_0)(x-x_0) + f_y(x_0,y_0)(y-y_0) \\
&+ \frac{1}{2!}(f_{xx}(x_0,y_0)(x-x_0)^2 \\
&+ f_{xy}(x_0,y_0)(x-x_0)(y-y_0) \\
&+ f_{yy}(x_0,y_0)(y-y_0)^2) + \cdots
\end{aligned}
$$

Taylor Series provides basis for all error analysis.

Inertial Error Dynamics

From Strapdown Equations to Error Equations

Inertial Error Dynamics

From Strapdown Equations to Error Equations

- Nav equations require perfect inertial instruments

Inertial Error Dynamics

From Strapdown Equations to Error Equations

- Nav equations require perfect inertial instruments
- Instruments are not perfect

Inertial Error Dynamics

From Strapdown Equations to Error Equations

- Nav equations require perfect inertial instruments
- Instruments are not perfect
- Inertial instrument error sources (bias, scale factor errors, etc.) generate erroneous rate and acceleration inputs to the nav equations

Inertial Error Dynamics

From Strapdown Equations to Error Equations

- Nav equations require perfect inertial instruments
- Instruments are not perfect
- Inertial instrument error sources (bias, scale factor errors, etc.) generate erroneous rate and acceleration inputs to the nav equations
- Need to figure out the effects on the outputs of these equations

Inertial Error Dynamics

Strapdown Error Equations

Inertial Error Dynamics

Strapdown Error Equations

- Nature of Errors

Inertial Error Dynamics

Strapdown Error Equations

- Nature of Errors
- Error Operator

Inertial Error Dynamics

Strapdown Error Equations

- Nature of Errors
- Error Operator
- Error Equations

Inertial Error Dynamics

Strapdown Error Equations: Nature of Errors

Inertial Error Dynamics

Strapdown Error Equations: Nature of Errors

- Errors propagate according to derived equations

Inertial Error Dynamics

Strapdown Error Equations: Nature of Errors

- Errors propagate according to derived equations
- Errors are generally small, so approximating assumptions help simplify the math

Inertial Error Dynamics

Strapdown Error Equations: Nature of Errors

- Errors propagate according to derived equations
- Errors are generally small, so approximating assumptions help simplify the math
  - linearize otherwise non-linear dynamics

Inertial Error Dynamics

Strapdown Error Equations: Nature of Errors

- Errors propagate according to derived equations
- Errors are generally small, so approximating assumptions help simplify the math
  - linearize otherwise non-linear dynamics
  - small angle assumptions: $\sin\theta \approx \theta$

Inertial Error Dynamics

Strapdown Error Equations: Error Operator

Inertial Error Dynamics

Strapdown Error Equations: Error Operator

- If A is any quantity, we define error operator as follows:

$$\delta A = \hat{A} - A, \tag{18}$$

where A is the true quantity of interest and $\hat{A}$ is an apparent quantity. The latter could be a measured value, and estimated value, or some computed value. This means that

$$\hat{A} = A + \delta A$$

i.e., the apparent value is the true value plus some error.

Inertial Error Dynamics

Strapdown Error Equations: Error Operator

Inertial Error Dynamics

Strapdown Error Equations: Error Operator

- Error operator satisfies:

$$\delta(AB) = \delta A \cdot B + A \cdot \delta B \tag{19}$$

Inertial Error Dynamics

Strapdown Error Equations: Error Operator

- Error operator satisfies:

$$\delta(AB) = \delta A \cdot B + A \cdot \delta B \tag{19}$$

- Proof:

$$\begin{aligned}
\delta(AB) &= \widehat{AB} - AB \\
&= \hat{A}\hat{B} - AB \\
&= (A + \delta A)(B + \delta B) - AB \\
&= AB + A \cdot \delta B + \delta A \cdot B + \delta A \cdot \delta B - AB \\
&= \delta A \cdot B + A \cdot \delta B
\end{aligned}$$

since we may assume that $\delta A \cdot \delta B \approx 0$.

Inertial Error Dynamics

Strapdown Error Equations

Inertial Error Dynamics

Strapdown Error Equations

- Applying these to the three differential equations, yields three error equations:

$$\delta \dot{C}_B^L = \delta C_B^L \Omega_{IB}^B + C_B^L \delta \Omega_{IB}^B - \left(\delta \Omega_{IE}^L + \delta \Omega_{EL}^L\right) C_B^L - \left(\Omega_{IE}^L + \Omega_{EL}^L\right) \delta C_B^L \tag{20}$$

$$\delta \dot{v}^L = \delta C_B^L a_{sf}^B + C_B^L \delta a_{sf}^B + \delta g' - \left(2\delta \Omega_{IE}^L + \delta \Omega_{EL}^L\right) v^L - \left(2\Omega_{IE}^L + \Omega_{EL}^L\right) \delta v \tag{21}$$

$$\delta \dot{C}_L^E = \delta C_L^E \Omega_{EL}^L + C_L^E \delta \Omega_{EL}^L \tag{22}$$

Inertial Error Dynamics

Strapdown Error Equations

Inertial Error Dynamics

Strapdown Error Equations

- Introducing **small angular rotations** for tilt and position

Inertial Error Dynamics

Strapdown Error Equations

- Introducing **small angular rotations** for tilt and position
  - $\widehat{C}_B^L$, the computed (i.e., apparent) version of C_B^L is related additively to the true version by the error δC_B^L

$$\widehat{C}_B^L = C_B^L + \delta C_B^L$$

Inertial Error Dynamics

Strapdown Error Equations

- Introducing **small angular rotations** for tilt and position
  - $\widehat{C}_B^L$, the computed (i.e., apparent) version of C_B^L is related additively to the true version by the error δC_B^L

$$\widehat{C}_B^L = C_B^L + \delta C_B^L$$

  - It is also related to C_B^L by a small direction cosine matrix relating the local level frame to the computer frame $\widehat{L}$:

$$\widehat{C}_B^L = C_{\widehat{L}}^L C_B^L$$

Inertial Error Dynamics

Strapdown Error Equations

Inertial Error Dynamics

Strapdown Error Equations

- Substituting the second equation into the first allows us to solve for δC_B^L:

$$\begin{aligned}\delta C_B^L &= \widehat{C}_B^L - C_B^L \\ &= \widehat{C_L^L} C_B^L - C_B^L \\ &= (\widehat{C_L^L} - I) C_B^L\end{aligned}$$

Now, let's introduce $\Phi = I - \widehat{C_L^L}$ to obtain:

$$\delta C_B^L = -\Phi C_B^L$$

Inertial Error Dynamics

Strapdown Error Equations

Inertial Error Dynamics

Strapdown Error Equations

- Using small angle approximations, $C_L^{\widehat{L}}$ can be written in the form

$$C_L^{\widehat{L}} = \begin{bmatrix} 1 & \phi_z & -\phi_y \\ -\phi_z & 1 & \phi_x \\ \phi_y & -\phi_x & 1 \end{bmatrix}.$$

Thus, $\Phi = I - C_L^{\widehat{L}}$ is seen to be a skew-symmetric rotation matrix:

$$\Phi = \begin{bmatrix} 0 & -\phi_z & \phi_y \\ \phi_z & 0 & -\phi_x \\ -\phi_y & \phi_x & 0 \end{bmatrix}.$$

Inertial Error Dynamics

Strapdown Error Equations

Inertial Error Dynamics

Strapdown Error Equations

- Φ is therefore equal to the cross product operator associated with a vector $\phi^L = \begin{bmatrix} \phi_x^L & \phi_y^L & \psi_z^L \end{bmatrix}^\top$. I.e.,

$$\Phi = \left[\phi^L \times \right]$$

Inertial Error Dynamics

Strapdown Error Equations

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- The three components of ϕ^L will become three states in our inertial error dynamics Kalman filter.

Inertial Error Dynamics

Strapdown Error Equations

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$$\Phi = \left[\phi^L \times \right]$$

- The three components of ϕ^L will become three states in our inertial error dynamics Kalman filter.
- ϕ_x and ϕ_y are called tilts, while ψ_z is called the azimuth error.

Inertial Error Dynamics

Strapdown Error Equations

Inertial Error Dynamics

Strapdown Error Equations

- By a similar derivation, we can introduce a small angular position angle error $e^L = \begin{bmatrix} e_x^L & e_y^L & e_z^L \end{bmatrix}^\top$

Inertial Error Dynamics

Strapdown Error Equations

- By a similar derivation, we can introduce a small angular position angle error $e^L = \begin{bmatrix} e_x^L & e_y^L & e_z^L \end{bmatrix}^\top$
- e^L is associated with δC_E^L just as ϕ^L is associated with δC_B^L:

$$\delta C_E^L = -E^L C_E^L,$$

where E^L is the skew-symmetric matrix form of e^L.

Inertial Error Dynamics

Strapdown Error Equations

- By a similar derivation, we can introduce a small angular position angle error $e^L = \begin{bmatrix} e_x^L & e_y^L & e_z^L \end{bmatrix}^\top$
- e^L is associated with δC_E^L just as ϕ^L is associated with δC_B^L:

$$\delta C_E^L = -E^L C_E^L,$$

  where E^L is the skew-symmetric matrix form of e^L.
- The three components of e^L yield the three position error states in a typical navigation Kalman filter.

Inertial Error Dynamics

Strapdown Error Equations

Inertial Error Dynamics

Strapdown Error Equations

- We are working to simplify Eq. (20):

$$\delta \dot{C}_B^L = \delta C_B^L \Omega_{IB}^B + C_B^L \delta \Omega_{IB}^B - \left(\delta \Omega_{IE}^L + \delta \Omega_{EL}^L \right) C_B^L - \left(\Omega_{IE}^L + \Omega_{EL}^L \right) \delta C_B^L$$

Inertial Error Dynamics

Strapdown Error Equations

- We are working to simplify Eq. (20):

$$\delta \dot{C}_B^L = \delta C_B^L \Omega_{IB}^B + C_B^L \delta \Omega_{IB}^B - \left(\delta \Omega_{IE}^L + \delta \Omega_{EL}^L \right) C_B^L - \left(\Omega_{IE}^L + \Omega_{EL}^L \right) \delta C_B^L$$

- We have an expression for δC_B^L.

Inertial Error Dynamics

Strapdown Error Equations

Inertial Error Dynamics

Strapdown Error Equations

- Consider the earth rate error term $\delta\Omega_{IE}^{L}$. Since $\Omega_{IE}^{L} = C_{E}^{L}\Omega_{IE}^{E}$ and $\delta\Omega_{IE}^{E} = 0$, we derive:

$$\delta\Omega_{IE}^{L} = \delta C_{E}^{L}\Omega_{IE}^{E}.$$

Inertial Error Dynamics

Strapdown Error Equations

- Consider the earth rate error term $\delta\Omega_{IE}^{L}$. Since $\Omega_{IE}^{L} = C_{E}^{L}\Omega_{IE}^{E}$ and $\delta\Omega_{IE}^{E} = 0$, we derive:

$$\delta\Omega_{IE}^{L} = \delta C_{E}^{L}\Omega_{IE}^{E}.$$

- But $\delta C_{E}^{L} = -E^{L}C_{E}^{L}$. Therefore,

$$\begin{aligned}\delta\Omega_{IE}^{L} &= -E^{L}C_{E}^{L}\Omega_{IE}^{E} \\ &= -E^{L}\Omega_{IE}^{L}.\end{aligned}$$

Inertial Error Dynamics

Strapdown Error Equations

Inertial Error Dynamics

Strapdown Error Equations

- For the error in transport rate, $\delta\Omega_{EL}^{L}$, using the error operator on Eq. (10), a new error terms having to do with altitude error shows up:

$$\begin{aligned}\delta\omega_{EL}^{L} &= \delta\left[\frac{1}{R}u_r^L \times v^L\right] \\ &= \frac{1}{R}u_r^L \times \delta v^L - \frac{\delta h}{R^2}\left[u_r^L \times v^L\right]\end{aligned}$$

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- Here, $R = R_e + h$ for a spherical earth of radius R_e.

Inertial Error Dynamics

Inertial Error Dynamics

- Eq. (20) can now be rewritten as:

$$\dot{\Phi}^L = -\left(\Omega_{IE}^L + \Omega_{EL}^L\right)\Phi^L - C_B^L \delta\Omega_{IB}^B + \Omega_{IE}^L E^L + \delta\Omega_{EL}^L \quad (23)$$

where $\delta\Omega_{EL}^L$ is the skew-symmetric matrix form of

$$\delta\omega_{EL}^L = \frac{1}{R}\left(u_R^L \times \delta v^L\right) - \frac{\delta h}{R^2}\left(u_R^L \times v^L\right)$$

Inertial Error Dynamics

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- Similarly, the velocity equation can be rewritten:

$$\delta\dot{v}^L = C_B^L \delta a_{sf}^B + a_{sf}^B \times \phi^L - \left(2\omega_{IE}^L + \omega_{EL}^L\right) \times \delta v^L + \\ - \left(\delta\omega_{EL}^L - 2e^L \times \omega_{IE}^L\right) \times v^L + \frac{2g\delta h}{R_e} u_R^L + \delta g^L$$

while the equation for the position angle error e^L itself is

$$\dot{e}^L = -\Omega_{EL}^L e^L + \delta\omega_{EL}^L \tag{24}$$

Inertial Error Dynamics

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Strapdown Error Equations

- Further Simplifications Using Telescope Pointing Angle Error

Inertial Error Dynamics

Strapdown Error Equations

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- $\psi^L = \phi^L - e^L$

Inertial Error Dynamics

Strapdown Error Equations

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- Leads to:

Inertial Error Dynamics

Strapdown Error Equations

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- Leads to:

$$\delta \dot{R}^L = \delta v^L - \Omega^L_{EL} \delta R^L \tag{25}$$

and

$$\delta \dot{v}^L = C^L_B \delta a^B_{sf} + a^B_{sf} \times \psi^L - \frac{g}{R_e} \Delta R^L_H - \left(2\omega^L_{IE} + \omega^L_{EL}\right) \times \delta v^L +$$
$$- \left(\delta \omega^L_{EL} - 2e^L \times \omega^L_{IE}\right) \times v^L + \frac{2g\delta h}{R_e} u^L_R + \delta g^L \tag{26}$$

Inertial Error Dynamics

Strapdown Error Equations

Inertial Error Dynamics

Strapdown Error Equations

- Finally, the equations for the telescope pointing angle errors are:

Inertial Error Dynamics

Strapdown Error Equations

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Inertial Error Dynamics

Strapdown Error Equations

Inertial Error Dynamics

Strapdown Error Equations

- Three error equations yield three vector quantities of interest

Inertial Error Dynamics

Strapdown Error Equations

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  - velocity error vector (3) δV^L (or δv^L)

Inertial Error Dynamics

Strapdown Error Equations

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  - position error vector (3) δR derived from δC_L^E
  - velocity error vector (3) δV^L (or δv^L)
  - attitude error vector (3) ϕ_x, ϕ_y, ψ_z from δC_B^L

Inertial Error Dynamics

Strapdown Error Equations

Inertial Error Dynamics

Strapdown Error Equations

- See Matlab / Simulink Models

Inertial Error Dynamics

Strapdown Error Equations

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  - Strapdown Navigation Equations: TBD

Inertial Error Dynamics

Strapdown Error Equations

- See Matlab / Simulink Models
  - Strapdown Navigation Equations: TBD
  - Strapdown Navigation Error Equations: TBD

Initialization: Alignment

- Coarse Leveling
  - Gimbal system limited by platform torquing rates
  - Strapdown systems have no real rate limits
  - Time average of accel outputs scaled by $1/g$
  - Yields third row of DCM to begin fine alignment
- Fine Alignment
- Alignment filter aids in further leveling

The Basic 9-State Kalman Filter

Topics

The Basic 9-State Kalman Filter

Topics

- Whole State Filters vs. Error State Filters

The Basic 9-State Kalman Filter

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- Whole State Filters vs. Error State Filters
- The Canonical Nine States

The Basic 9-State Kalman Filter

Topics

- Whole State Filters vs. Error State Filters
- The Canonical Nine States
- INS Error State Dynamics and Transition Matrix

9-state Filter

Whole State vs. Error State Filters

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  - model whole position, velocity, attitude
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  - are highly non-linear

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Whole State vs. Error State Filters

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  - remove some of the nonlinearities by differencing outputs from aiding sensors with corresponding quantities from INS
  - exhibit slowly varying dynamics: easier to linearize
  - have smaller quantities that allow us to take advantage of small angle assmptions

The Canonical Nine States

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- 3 attitude errors: ϕ_x, ϕ_y, ψ_z
  - ϕ_x, ϕ_y known as tilt errors
  - ψ_z known as azimuth error

Exercises

Exercises

- 1. Rewrite equation (20)

$$\delta \dot{C}_B^L = \delta C_B^L \Omega_{IB}^B + C_B^L \delta \Omega_{IB}^B - \left(\delta \Omega_{IE}^L + \delta \Omega_{EL}^L\right) C_B^L - \left(\Omega_{IE}^L + \Omega_{EL}^L\right) \delta C_B^L$$

to show the terms in the canonical order with position error, velocity error, and then attitude error.

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  to show the terms in the canonical order with position error, velocity error, and then attitude error.
- 2. Determine the 7th row of the system matrix F for the canonical 9-state error model. Use the cross-product operator matrix as required, e.g.,

$$\Omega_{IE}^L = \left[\, \omega_{IE}^L \times \,\right] = \begin{bmatrix} 0 & \left[\omega_{IE}^L\right]_z & -\left[\omega_{IE}^L\right]_y \\ -\left[\omega_{IE}^L\right]_z & 0 & \left[\omega_{IE}^L\right]_x \\ \left[\omega_{IE}^L\right]_y & -\left[\omega_{IE}^L\right]_x & 0 \end{bmatrix}.$$

  Introduce simplifying notation! E.g., set $\alpha = \omega_{IE}^L$ Hint: Use telescope pointint angle error equation (27)

Challenges of INS Filters

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- Nine States Are Not Enough

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- However, they don't help characterize dynamics of aiding sensor errors
- They don't help estimate aiding states such as Doppler bias and boresight errors
- Or GPS receiver clock errors
- They also don't help us to estimate inertial instrument drifts such as bias drift or scale factor errors

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- Position dependent items such as latitude are also involved

Non-Linearities: Extended Kalman Filter

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- And also choosing the right measurement scheduling
- Also, certain kinds of divergence can be the result of poor observability (more on this later)

Exercises

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- Use Eq. (25)

$$\delta \dot{R}^L = \delta v^L - \Omega^L_{EL} \delta R^L,$$

write the first three rows of the system dynamics matrix F in the continuous Kalman formulation:

$$\dot{x} = Fx + w$$

Precise Orbit Determination (POD): Two-Body Equation of Motion

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- Basic two-body equation of relative motion:

  Let r be the vector from mass m_1 to m_2. Then

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  where $\mu = G(m_1 + m_2)$ and G is the gravitational constant.

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- For earth, $\mu_E := 3.986004418 \times 10^{14}$ $\mathrm{m}^3/\mathrm{sec}^2$ (from WGS84)

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- Energy is also conserved (if we ignore internal, heat-generating forces)

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- Perturbation theory facilitates linearization with respect to reference trajectory

Precise Orbit Determination (POD): Gravitational Perturbations

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  - Corrects Newtonian formula given in Eq. (28)

Precise Orbit Determination (POD): Non-Gravitational Perturbations

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- Drag (appreciable for LEO satellites)

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Precise Orbit Determination (POD): Non-Gravitational Perturbations

- Drag (appreciable for LEO satellites)
- Solar Radiation Pressure (SRP)
- Earth Radiation Pressure (ERP), albedo
- Other minor effects: thermal, magnetic, ...

Linearization about a Nominal Trajectory: Taylor Series

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- Multivariable Taylor Series

Single Variable Taylor Series

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- If f is a real-valued continuous function with continuous derivatives of all orders in a neighborhood of a point x_0, then the value of f at any other point x in that neighborhood can be expressed as an infinite series as follows:

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where $D^k_{x,x_0} f := (x - x_0)^k \left. \frac{d^k f}{dx^k} \right|_{x_0}$.

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- x is a scalar variable

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- Note that x is also (in our applications) an implicit vector-valued function of time $x = x(t)$. But we are ignoring that for now.

Linearization

Linearization

- We can unify the multivariable and single variable notation with the following definition:

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$$D^k_{x,x_0} f := \begin{cases} (x - x_0)^k \dfrac{d^k f}{dx^k}\Big|_{x_0}, & \text{if } \dim(x) = 1; \\ \left(\displaystyle\sum_{i=1}^{n} (x_i - [x_0]_i) \frac{\partial f}{\partial x_i}\Big|_{x_0} \right)^k, & \text{if } \dim(x) > 1. \end{cases} \tag{31}$$

Precise Orbit Determination (POD): Linearization

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- $x(t) = F\left[t, t_0, x_0(t)\right]$
  $x = \left[x_1, x_2, x_3, \dot{x}_1, \dot{x}_2, \dot{x}_3, \gamma\right]^\top$

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  $x = \left[x_1, x_2, x_3, \dot{x}_1, \dot{x}_2, \dot{x}_3, \gamma\right]^\top$
- $\dot{x}(t) = \dot{x}_{\text{ref}}(t) + \left[\dfrac{\partial F(t)}{\partial \lambda}\right]_{\lambda = x_{\text{ref}}} \left[x(t) - x_{\text{ref}}(t)\right] + \cdots,$
  γ is an m-vector of constant parameters associated with perturbation forces, and the derivative is evaluated at a point on the reference trajectory

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  γ is an m-vector of constant parameters associated with perturbation forces, and the derivative is evaluated at a point on the reference trajectory
- Or, equivalently
  $\dot{x}(t) = F(x^*, t) + \left[\dfrac{\partial F(x,t)}{\partial x(t)}\right]^* \left[x(t) - x^*(t)\right] + \cdots$

General Linearization

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- Typically, we linearize about an equilibrium point x_0 or a reference (or nominal) function $x_{\text{ref}}(t)$

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- Typically, we linearize about an equilibrium point x_0 or a reference (or nominal) function $x_{\text{ref}}(t)$
- Equilibrium: $\dot{x} = f(x_0) = 0$, define $\delta x(t) = x(t) - x_0$. Then, since x_0 is a constant, we have
$$\frac{d\delta x(t)}{dt} = \frac{dx(t)}{dt} - \frac{dx_0}{dt} = \dot{x}(t) - 0 = \dot{x}(t)$$

General Linearization

- Typically, we linearize about an equilibrium point x_0 or a reference (or nominal) function $x_{\text{ref}}(t)$
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- Note: Equilibrium points are called critical points in mathematics (points where the gradient is zero)

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- Then the equation of motion becomes: $\dot{x}(t) = Ax(t)$, where $A(t) := \left[\frac{\partial \dot{X}(t)}{\partial X(t)}\right]^*$
- The state transition matrix Φ is associated with the homgeneous solution of the linearized differential equation above, and enables future state to be computed from the current state :

$$x(t) = \Phi(t, t_0)x(t_0)$$

and also itself satisfies the matrix differential equation

$$\dot{\Phi}(t, t_0) = A(t, t_0)\Phi(t, t_0),$$

with initial condition $\Phi(t_0, t_0) = I$

Simplified Perturbation Models

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- SGP: Simplified General Perturbations (orbital period ¡ 225 minutes)
- SDP: Simplified Deep Space Perturbations (orbital period ¿ 225 minutes)
- Many implementations available: Python, MATLAB, FORTRAN, C

Example: 2D, 7-state

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- Consider a simple, planar example: $X = \left[X_1, X_2, \dot{X}_1, \dot{X}_2, \mu, X_s, Y_s\right]^\top$, where X_1, X_2 are the two SV position coordinates and the γ (nuisance) parameters are μ and the monitor station coordinates X_s, Y_s.

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- Using $\ddot{X}_1 = -\mu X_1/r^3$ and $\ddot{X}_2 = -\mu X_2/r^3$, and assuming that μ and the station coordinates are constant, we obtain $\dot{X} = \left[\dot{X}_1, \dot{X}_2, -\frac{\mu X_1}{r^3}, -\frac{\mu X_2}{r^3}, 0, 0, 0\right]^\top$

Example: 2D, 7-state Model

Measurement Model

Example: 2D, 7-state Model

Measurement Model

- The A matrix becomes:

$$A = \begin{bmatrix} 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ -\frac{\mu}{r^3} + \frac{3\mu X_1^2}{r^5} & \frac{3\mu X_1 X_2}{r^5} & 0 & 0 & -\frac{X_1}{r^3} & 0 & 0 \\ \frac{3\mu X_1 X_2}{r^5} & -\frac{\mu}{r^3} + \frac{3\mu X_2^2}{r^5} & 0 & 0 & -\frac{X_2}{r^3} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}^*$$

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- If the A matrix were constant, we could easily calculate the state transition matrix Φ to be the matrix exponential of A:

$$\Phi(t - t_0) = \exp\left[A((t - t_0)\right]$$

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- We evaluate A at X_0
- We have to integrate the Φ differential equation from t_0 to t get the new value of the state transition matrix that will allow us to extrapolate the state from time t_0 to t, keeping it mind that X_1, X_2, r are functions of time.

Example: 2D 7-state Model

System Dynamics Matrix

$$A(t) = \begin{pmatrix} 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ \frac{\mu\left(2X^2-Y^2\right)}{(X^2+Y^2)^{5/2}} & \frac{3XY\mu}{(X^2+Y^2)^{5/2}} & 0 & 0 & -\frac{X}{(X^2+Y^2)^{3/2}} & 0 & 0 \\ \frac{3XY\mu}{(X^2+Y^2)^{5/2}} & -\frac{\mu\left(X^2-2Y^2\right)}{(X^2+Y^2)^{5/2}} & 0 & 0 & -\frac{Y}{(X^2+Y^2)^{3/2}} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

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$$H := \frac{\partial \rho}{\partial X} = \left[\frac{X - X_s}{\rho} \quad \frac{Y - Y_s}{\rho} \quad 0 \quad 0 \quad 0 \quad -\frac{X - X_s}{\rho} \quad -\frac{Y - Y_s}{\rho} \right]$$

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- If measurements arrive at different times after epoch, then they can be related back to epoch state by applying Φ: Let $H(i)$ be the H matrix at time $t_i > t_k$. Define new observation matrix Γ whose i^{th} element is the row block $H(i)\Phi(t_i, t_k)$

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- Stack all measurements via $z = \Gamma x(k) + v(k)$

GPS OCX

Orbit Integrator (OI)

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GPS OCX

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- Force models:

  $f = f_{\text{grav}} + f_{\text{3b}} + f_{\text{srp}} + f_{\text{tides}} + f_{\text{rel}} + f_{\text{earthrad}} + f_{\text{custom}}$ where
  f_{grav} = acceleration due to gravity
  f_{2b} = acceleration due to sun and/or moon f_{srp} = acceleration due to solar radiation pressure
  f_{tides} = acceleration due to to solid earth, pole, and ocean tides
  f_{rel} = acceleration due to relativity
  f_{earthrad} = acceleration due to earth albedo and IR radiation
  f_{custom} = acceleration due to custom force model (e.g., constant and once-per-rev accelerations)

  See gcore.pdf for details on the force models

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- To facilitate filter estimation of the states, we compute partial derviatives of the state with respect to $r_0, \dot{r}_0$ and p; partials represent sensitivity of satellite state to uncertainty in initial conditions (i.e., prior state)

GPS OCX

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where $A = \frac{\partial f}{\partial r}, B = \frac{\partial f}{\partial \dot{r}}$ and $C = \frac{\partial f}{\partial p}$

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GPS OCX

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- Foce model parameters read in from database
- ODE solver includes single-step (Runge-Kutta) and multi-step (Adams)

GPS OCX

RTGx Processing Flow

GPS OCX

RTGx Processing Flow

- 1. SRIF Filter

GPS OCX

RTGx Processing Flow

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- 2. UD Smoother with Ambiguity Resolution

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RTGx Processing Flow

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- 3. SRIF Filter to apply ambiguity constraints

GPS OCX

RTGx Processing Flow

- 1. SRIF Filter
- 2. UD Smoother with Ambiguity Resolution
- 3. SRIF Filter to apply ambiguity constraints
- 4. UD Smoother without Ambiguity Resolution

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RTGx Inputs

GPS OCX

RTGx Inputs

- 1. Parameters and options

GPS OCX

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- 2. Deterministic state mapping functions, which come from integrating spacecraft state variational equations

GPS OCX

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GPS OCX

RTGx Inputs

- 1. Parameters and options
- 2. Deterministic state mapping functions, which come from integrating spacecraft state variational equations
- 3. Partial derivatives of the measurements with respect to the problem parameters
- 4. A time sequence of stochastic events for any of the stochastic parameters. As well as noting white noise and random walk updates, the time sequence will indicate parameters added and removed

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RTGx Optional Outputs

GPS OCX

RTGx Optional Outputs

- 1. Estimates of linear combinations of state variables

GPS OCX

RTGx Optional Outputs

- 1. Estimates of linear combinations of state variables
- 2. Sigmas of linear combinations of state variables

GPS OCX

RTGx Optional Outputs

- 1. Estimates of linear combinations of state variables
- 2. Sigmas of linear combinations of state variables
- 3. SRI matrix

GPS OCX

RTGx Optional Outputs

- 1. Estimates of linear combinations of state variables
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- 3. SRI matrix
- 4. Covariance matrix

GPS OCX

RTGx Optional Outputs

- 1. Estimates of linear combinations of state variables
- 2. Sigmas of linear combinations of state variables
- 3. SRI matrix
- 4. Covariance matrix
- 5. Sensitivity information

GPS OCX

RTGx Optional Outputs

- 1. Estimates of linear combinations of state variables
- 2. Sigmas of linear combinations of state variables
- 3. SRI matrix
- 4. Covariance matrix
- 5. Sensitivity information
- 6. Smoothing/forward smoothing/removal smoothing coefficients